Superseding
MIL-H-5166C(USAF)
15 October 1969

MILITARY SPECIFICATION

MANUAL, TECHNICAL: ASSEMBLY, SERVICING, AND FIELD MAINTENANCE INSTRUCTIONS, WITH ILLUSTRATED PARTS BREAKDOWN

(For Guided Missiles and Pilotless Aircraft)

This specification is approved for use by Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers the preparation of assembly, servicing, and field maintenance instructions for guided missiles and pilotless aircraft. Guided missiles and pilotless aircraft include guided bombs, guided rockets, guided aircraft rocket interceptors, air-to-air missiles, air-to-surface missiles, and pilotless spacecraft.

NOTE: Field maintenance includes organizational, intermediate and squadron level Precision Measurement Equipment Laboratory.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-38784	Manuals, Technical: General Style and Format Requirements

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC/LOLME, WPAFB, OH, 45433, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

MIL-P-38790	Printing Production of Technical Manuals: General Requirements for
MIL-M-38807	Manuals, Technical: Illustrated Parts Breakdown, Preparation of

STANDARDS

Military

MIL-STD-838	Lubrication of Military Equipment
MIL-STD-875	Type Designation for Aeronautical and Support Equipment

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

AIR FORCE REGULATION

AFR 127-100	Explosive Safety Standards

TECHNICAL ORDERS

TO 00-85-Series	Protective Packing and Preservation Packing - General

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preparation. The general manner of preparation of text, illustrations, diagrams, warnings, cautions, notes, health hazards, negatives, etc., shall be in accordance with MIL-M-38784, MIL-P-38790 and MIL-M-38807. The manual shall contain all information, general and specific, required by maintenance organizations, with the tools and test equipment authorized. The manual shall not cover maintenance which requires tools, test equipment, manpower, skills, etc., for depot maintenance, or authorized for depot organizations only.

3.2 Contents. The instructions shall provide all essential information required by personnel for performing maintenance on the components of the equipment. The instructions shall show the interface between the components and the end item equipment. Maintenance instructions shall enable personnel to test, troubleshoot, repair, and return to an acceptable performance standard, all components of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

equipment in a minimum of time with the tools, test equipment, and spare parts authorized for maintenance activities. Instructions shall cover all items comprising the equipment, such as accessories, components, wiring, junction boxes, etc. Procedures requiring a team of specialists for the accomplishment of maintenance tasks shall have the tasks of these individuals designated separately in sequence of operation.

3.2.1 Development of text. The requirements of MIL-M-38784 are applicable. Repetitious information shall be avoided except for clarity or where considerable maintenance time would be saved by repeating information to preclude excessive cross-referencing between manuals. Emphasis shall be placed on the specific steps of the maintenance procedures to be followed, the results which may be expected or desired, and the corrective measures required when such results are not obtained.

3.2.2 Environmental control. Environmental control information shall be provided. The manual shall include information on any special maintenance required under extreme temperature, altitude, and humidity conditions within the limits established by the design specification for the equipment.

3.2.3 Nomenclature. Nomenclature shall be in accordance with the requirements of MIL-M-38784.

3.2.4 Reference to parts of the equipment. Reference to parts of the equipent, and its components, etc., shall be by type designation, standard nomenclature, reference designation, or figure and index number, as applicable.

3.2.5 Abbreviations. Abbreviations shall be in accordance with MIL-M-38784.

3.2.6 Illustrations. Illustrations and diagrams shall be provided in accordance with MIL-M-38784.

3.2.6.1 Illustration keys. Legends, tables or text applicable to partial page illustrations shall, whenever possible, appear on the same page. For a full page illustration, the information shall appear on a facing page or the illustration, legend, table, and text may be arranged as a foldout page. If the legend, table, or text exceeds the allowable space, it may be continued on a second page

3.2.6.2 Reference designations. Electrical and electronic reference designations shall be in accordance with MIL-M-38784.

3.2.6.3 Diagrams. The manual shall include diagrams and associated data necessary for maintenance personnel to: (a) understand the make-up and function of each intercomponent, power, control, and signal circuit, and of each mechanical system; (b) trace each circuit and mechanical system; (c) accomplish general and specific troubleshooting on inoperative or malfunctioning circuits or mechanical systems. Foldout diagrams shall be in accordance with MIL-M-38784.

3.2.6.3.1 Explanatory text. An explanation of the diagrams used in the manual shall be given as to their content and proper use. This shall include the working relationship to material in other sections, how to locate a particular diagram, an explanation of the

Source: https://assist.dla.mil -- Downloaded: 202610-17T04:23Z
Check the source to verify that this is the current version before use.

wiring identification system used on the diagrams, and any other information considered essential for proper understanding and use.

3.2.6.3.2 Symbols. The requirements of MIL-M-38784 are applicable.

3.2.6.3.3 Schematic diagram general requirements. Where the equipment's nature, relative complexity, and number of components make it practicable, a single schematic diagram shall be prepared to cover all signal circuits, power circuits, and control circuits. When a single diagram is impracticable, a separate diagram shall be prepared for each item of equipment. Any circuits left in "mid-air" and picked up on other diagrams and which terminate with a plug, terminal strip, jack, etc., shall show a composite plug grouping, and each lead shall be extended through the plug, terminal strip, jack, etc., and labeled to indicate the necessary designator of the first connection within the other component, or other significant point of termination.

3.2.6.3.3.1 Interconnections. When the equipment obtains information from, or provides information to other items of equipment of a system, diagrams shall be provided showing all required interconnections.

3.2.6.3.3.2 Circuits. Power circuits, control circuits and signal flow within a particular circuit(s) or functional loop(s) shall be identified. Any use of color shall be in accordance with MIL-M-38784.

3.2.6.3.4 Schematic diagrams for equipment. A schematic diagram of an item of equipment shall show the following:

a. Subassemblies shall be indicated by a series of broken lines.

b. For each detail part, identification shall be by reference designations.

c. For each detail part, the principal electrical characteristics expressed in actual values (ohms, microfarads, microhenrys, etc.), shall be tabulated next to the diagram and indexed by reference designation to applicable points on the diagram when sufficient space is not available on the diagram itself. Values normally shall be expressed as numbers, omitting the unit of measurement (such as ohms). However, a general note shall state which units are used.

d. Wave shapes, duration, and amplitude of voltages at designated test points throughout the circuit and at interconnecting points within the equipment or systems, under normal test conditions and other specified conditions shall be indicated on the schematic diagram of the component.

3.2.6.3.5 Overall schematic diagrams. If separate component diagrams are not provided, the overall equipment schematic shall show the information called for in 3.2.6.3.4., and in addition thereto, interequipment circuits showing the connections and terminals properly numbered, with a series of broken lines indicating items of equipment.

Source: https://assist.dla.mil -- Downloaded: 202640-17T04:23Z
Check the source to verify that this is the currentversion before use.

3.2.6.3.6 Power and control schematics. Separate schematic diagrams shall be prepared, as follows, to show the power and control circuits when it is impracticable to show such information on the overall schematic diagram: each power circuit and each control circuit with its relationship to the signal circuits using a combination of blocks, schematics, or pictorial diagrams; tolerances of power distribution associated with each power and control circuit.

3.2.6.3.6.1 Logic diagrams. When appropriate to the design of the equipment, overall operation of the equipment shall be shown by means of conventional logic diagrams. Current generation of equipment utilizing solid state and integrated circuit devices is understood and troubleshooting is accomplished by means of logic diagrams rather than detailed schematics.

3.2.6.3.7 Waveform diagrams. If not provided on functional and schematic diagrams and required in support of maintenance tasks, these diagrams shall show the waveforms and nominal values at designated points (actual photographs, where possible) for normally functioning equipments, or systems as seen in an oscilloscope of a type listed in Section II of the manual. Operating conditions and switch positions of the operating equipment shall be noted at the time of each indication.

3.2.6.3.8 Interconnection cable diagrams. These diagrams shall show the cabling or connections for the system, equipment, or component. Interconnections between components or equipments shall be indicated by either a diagram or a table.

3.2.6.3.9 Combination diagrams. Combinations of the information required by 3.2.6.3.7 and 3.2.6.3.8 may be shown on a single diagram, where practicable.

3.2.6.3.10 Module schematic diagrams. Schematic diagrams of electronic modules, when shown separately, shall be arranged to indicate signal flow progressing from left to right. Where possible, an equivalent block/logic diagram of the module circuitry shall be placed on the schematic to aid in interpreting the schematic.

3.2.6.4 Test point identification. Test point identification shall be incorporated on schematic wiring diagrams, tables, charts and illustrations. Test point identification will not be required for equipment with self-test capability, equipment designed to be tested by an automatic checkout system, and equipment designed with critical test points accessible at front panel connectors. Where the design of the equipment includes built in test capability, tests performed using this capability will be identified but not outlined in detail. Any function requiring the participation of maintenance personnel, or interpretation of test results, shall be included in detail. Instructions required to determine the operability of the design built in test circuits and components, fault isolation and repair instructions, shall be included. Symbols shall be as marked on the equipment, or as they appear on applicable engineering drawings. Voltages, waveforms, etc., should be placed adjacent to the test point and if required may be in tabular form. Indiscriminate use of test points shall be avoided; for example, test point identification shall normally be restricted to signal tracing test points and voltage and continuity check points. The following test point identification system is commonly used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

3.2.6.4.1 Major test points. A star-enclosed Arabic numeral, ✯ , is used to identify and designate the test points used in checking the overall functions of, and localizing trouble to individual components, or groups of components. Major test points are identified on schematic wiring diagrams, tables, charts, and illustrations by use of the Arabic numerals, 1, 2, 3, etc., enclosed in a star; but will be referred to in the text as test point 1, test point 2, etc. Typical examples of major test points are: (a) the input terminals for the supply source voltage; (b) the high voltage output terminals of the equipments power system; (c) the signal input terminals of the equipment; (d) the signal output terminals of the equipment; and (e) the major points of power or voltage distribution within the equipment.

3.2.6.4.2 Secondary test points. An encircled capital letter, (A) , is used to identify and designate the test points used in isolating causes of subnormal performance within a specific assembly or subassembly of the equipment. Secondary test points are identified on schematic wiring diagrams, tables, charts, and illustrations by use of capital letters, A, B, C through Z; AA, AB through AZ; BA, BB through BZ, etc., enclosed in a circle; but will be referred to in the text as test point A, test point B, etc. Typical examples of secondary test points are: (a) circuit supply voltage terminals; (b) signal output points used in checking the gain of a stage or group of stages.

3.2.6.4.3 Minor test points. An encirled capital letter and Arabic numeral, (A1) , is used to identify and designate the test points used in isolating causes of abnormal indications within a specific circuit of the equipment. Minor test points are identified on schematic wiring diagrams, tables, charts and illustrations by use of a capital letter and Arabic numeral, A1, A2, etc., enclosed in a circle but will be referred to in the test as test point A1, test point A2, etc. Typical examples of minor test points to be so designated are: (a) specific voltage and resistance check points; (b) all other points used in checking for trouble in a specific circuit. Different letters are assigned to each component.

3.2.6.4.4 Assignment of symbols. Test point identification symbols on the schematic wiring diagrams shall be assigned consecutively throughout the composite equipment. Once assigned, a symbol shall not be reassigned to another test point. An explanation of the use of test points shall appear in the Foreword of the manual.

3.2.7 Cable fabrication. When cable assemblies are not supplied, but are required for bench test set-up, instructions shall be provided for fabricating interconnecting cable assemblies from spares and bulk supplies. The part number, drawing number, and manufacturer or designer for each part of the cable assembly shall be shown, as well as identification of wires, connectors, pin connections, and letters or other designators.

3.2.8 Warhead data. The manual shall include applicable instructions for "high explosive" and "incendiary" type warheads.

3.3 Arrangement. Manuals shall normally be arranged in the following manner:

Front Matter

Source: https://assist.dla.mil -- Downloaded: 202610-17T04:23Z
Check the source to verify that this is the current version before use.

Chapter 1	Description
Chapter 2	Special Tools, Test Equipment, and List of Consumables
Chapter 3	Unpacking, Assembly, Checkout, Ground Handling and Servicing
Chapter 4	Inspection Requirements
Chapter 5	Maintenance
Chapter 6	Packing, Shipping and Storage
Chapter 7	Contingency Procedures
Chapter 8	Training Missiles
Chapter 9	Illustrated Parts Breakdown
Chapter 10	Difference Data

NOTE: This chapter alignment applies to present day missile concepts. Other systems may realign chapters to meet needs of their particular requirements.

3.3.1 Front matter. The title page, list of effective pages, foreword, table of contents, list of illustrations, list of tables, and safety summary shall be prepared per MIL-M-38784.

3.3.1.1 Safety summary. This information shall precede chapter I. Data shall be prepared to describe all major hazards to personnel and equipment safety that are peculiar to the tasks covered. Tabular form presentation is preferred.

3.3.1.2 Safety and accident prevention. The safety summary will describe the safety requirements and accident prevention precautions applicable to the missile, including cleaning and painting hazards, fire minimum safety zones, explosive hazards, and any other data applicable.

3.4 Chapter I, description. This chapter shall give a brief description of the missile, including weight and dimensions. Dimensions shall be given in tabular format similar to table I. The principles of operation shall be described, including the interface with the aircraft launcher.

3.4.1 Principles of operation. The principles of operation of the missile shall be described. This shall be in enough detail to provide an understanding of the missile so that logical effective troubleshooting may be performed in support of the established maintenance concept. Only that information which will be of assistance to maintenance personnel in the performance of their tasks shall be presented. Special emphasis shall be given to features that will affect recognition and operation of the missile. This data may be given in tabular format.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

TABLE 1. Leading particulars.

Launch Aircraft	_____
Launcher	_____
Guided Missile	_____
Length	_____
Diameter	_____
Wing Span, Installed	_____
Center of Gravity	_____
Design Gross Weight	_____
Electrical Power	
External Input from Aircraft	_____
Internal Input	_____
Target Detector Group/Fuzing System	_____
Diameter	_____
Length	_____
Weight	_____
Hydraulic, Source Pressure	_____
Type; IR, Contact Etc.	_____
Warhead	_____
Length	_____
Diameter	_____

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

TABLE I. Leading particulars - (Continued).

Warhead (cont)

- Weight, Nominal WT. _____
- Explosive WT. and Type (PBX, HBX, etc) _____
- Type (Frag, Rod or Blast, etc.) _____

Flt Control Group _____

- Length _____
- Diameter _____
- Weight _____
- Type, IR, Radar, etc. _____

Wing Assy _____

- Height _____
- Length/Width _____
- Weight _____
- Base _____
- Type (High, Low, Mid) _____

Rocket Motor

- Type: Solid, Liquid, etc. _____
- Length, Overall _____
- Ignition System _____
- Diameter _____
  - Fwd End _____
  - Aft Tapered End _____
  - Weight _____
  - Center of gravity, fwd or aft end _____

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

3.4.2 Systems and major components. The operation, functions, locations, and interrelationships of the systems and major components of the missile shall be described; for example:

a. Power plant. The power plant and all controls employed in normal operation.

b. Propulsion system. The complete propulsion system, including the type, specification, and quantity of propellant used, also the method of arming the system if an electro/mechanical safe and arm device is employed.

c. Power supply systems. The power supply for flight operation, the power supply for ground operation, and the interface between them. Where there are two power supply systems to be covered, they shall be distinguished by describing them respectively under the headings "Power Supply System (Flight Operation)" and "Power Supply System (Ground Operation)."

d. Flight surfaces. Flight surfaces, their functions and methods of operation; the automatic features of the system and the functions of any controls which may be used for manual operation; and the gyro control system. Limits of the movement of control surfaces shall be given in degrees from a reference mark on the airframe.

e. Flight control system. Any controls which are used to limit flight; for example, attitude (pitch, roll and yaw) electronic, mechanical, and aerodynamic force limits.

f. Guidance system. The guidance system and the interface with airborne and ground control systems.

g. Warhead section; type of warhead, blast, fragmentation, etc., and methods of initiation.

h. Target detection/fuze system: Type, description, etc.

i. System functional check. Any system used solely for the purpose of checking the operation of the missile. A description of all controls, receptacles, and equipment used for this purpose shall be included. Method of arming the system, if electro/mechanical safe and arm devices are utilized.

j. Firing system. All equipment and controls employed in firing the missile, including all SAFE/ARM and FUZE subsystem circuits external to the warhead. Information to prevent inadvertent firing shall be included.

k. Launching devices. Type of launching device which can be used. A general concept of the launching device(s), whether stationary or mobile, overall dimensions, whether adjustable in evaluation or azimuth, and type of launching (cart, rails, etc). All controls employed in the operation of the launching device.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

3.4.3 Illustrations. The following are desirable illustrations that shall be included in chapter I, with additions and omissions as applicable to a particular missile. Include a matrix of the missile components including training items.

a. A view of the complete missile taken from an angle which will most clearly portray the missile.

b. A three dimensional (top, side, and front) line drawing of the missile showing principle overall dimensions. It shall show the missile in flying position and at rest.

c. A rear view of the missile.

d. Station and frame diagrams, showing fuselage, wings, etc. This diagram shall indicate the dimensions of each station from the reference or datum line. A note explaining how station locations may be determined by measurement from a readily accessible point shall be included. The location of station zero shall be clearly indicated on all such diagrams. The diagram shall include missile Center of Gravity (CG) at launch and motor burnout.

e. Wing, rudder, and fin folding and tieback illustrations. Identification of fairing and locations shall be included.

f. Interior arrangement of compartments, defining if the component interfaces are joined by threads, clamps, etc.

3.5 Chapter 2, special tools, test equipment and list of consumables. This chapter shall list and illustrate the special tools and test equipment required for the work described. Such items shall be selected from the list of tools and test equipment approved by the Government by means of support equipment (SE) listings, provisioning conferences, Engineering Change Proposals (ECPs), and procurement documents. Standard types of tools, such as screwdrivers and pliers shall not be listed. Standard types of test equipments, such as voltmeters and tube testers shall be listed. When a contractor cannot obtain an approved list, a manual shall show the contractor's recommended special tools and test equipment. However, these recommendations shall be changed promptly, if necessary, to conform to the official Government list as soon as such information is provided to the contractor.

3.5.1 Special tools list. The list of special tools shall be arranged in the following format. Tools not illustrated in use in other chapters of the manual shall be illustrated following the Special Tools List in this chapter.

SPECIAL TOOLS LIST

Part (Tool) Number	Figure & Index No.	Nomenclature	Use

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

a. Part (tool) number. Tools shall be listed in alpha-numeric sequence by part (tool) number and manufacturer's five digit code or manufacturer's name and address.

b. Figure and index number. The figure and index numbers that appear in other chapters, or this chapter, of the manual, shall be shown.

c. Nomenclature. Nomenclature shall be in accordance with the requirements of MIL-M-38784.

d. Use. The purpose of each tool shall be stated. If the use is described in any part of the manual, reference to the applicable paragraph will suffice.

3.5.2 Test equipment list. The list of test equipment shall be arranged in the following format. Test equipment not illustrated in use in other chapters of the manual shall be illustrated following the Test Equipment List in this chapter (standard test equipment excepted).

TEST EQUIPMENT LIST

Type Designation	Alternate Type Designation	Figure & Index No.	Nomenclature	Use

a. Type designation. Test equipment shall be listed in alpha-numeric sequence by type designation, using the AN type designation, if assigned and the manufacturer's five digit code or manufacturer's name and address.

b. Alternate type designation. If an alternate item of test equipment can be used, its AN type designation, if assigned, or the commercial or manufacturer's designation shall be listed.

c. Figure and index number. The figure and index numbers that appear in other chapters, or this chapter, of the manual, shall be shown.

d. Nomenclature. Nomenclature shall be in accordance with the requirements of MIL-M-38784.

e. Use. The purpose of each item of test equipment shall be stated. If the use is described in any part of the manual, reference to the applicable paragraph will suffice.

3.5.3 Consumables. Expendable items and support materials (consumables) shall be listed. Sealants, lubricants, gaskets, seals, cleaning solvents, paint, etc., are considered expendable items. The list will be in tabular form, in alphabetical order by nomenclature, as follows:

Nomenclature	Specification(s)	Part number and FSCM

Source: https://assist.dla.mil -- Downloaded: 202610-17T04:23Z
Check the source to verify that this is the current version before use.

3.5.4 Missile containers. Containers shall be listed in tabular format giving: Figure No, container application and part No., No. of units contained, dimensions, cube, weight loaded, weight empty and any other information required. Illustrations shall follow the table.

3.6 Chapter 3, unpacking, assembly, checkout, ground handling and servicing. This chapter shall contain procedures for preparing the handling equipment for use, unpacking and inspecting missile components, operational test of the missile guidance set and assembly of the missile.

3.6.1 Unpacking. Observe all safety requirements. Unpacking explosive components, correct markings, and retaining serviceable packaging material for possible reuse, and handling empty containers and their dunnage. Include instructions for handling damaged containers; repair or disposition, also procedures for dropping components, criteria, etc.

3.6.2 Assembly. Provide essential instructions and illustrations for:

a. Removal of the missile and components from containers.

b. Assembly of the missile, including the motor/booster units. Instructions for installation of the engine shall be provided. The center of gravity (CG) with the engine removed and installed shall be stated; weight and balance data shall be included. For assembly, rigging, and adjustment instructions, the extent of coverage shall depend upon the complexity of the required operations. The instructions shall be presented in a style that will allow the task to be carried out with a minimum of interruptions and stoppages.

3.6.3 Inspection and checkout. Inspection and checkout shall be performed on specified components, assemblies and ready missiles as required.

3.6.3.1 Inspection. All missile components, sections etc., repaired or replaced shall be inspected before and after installation per existing instructions. Provide the repair and inspection criteria.

3.6.3.1.1 Drop criteria. Missile component drop criteria shall be in the following format:

MISSILE COMPONENT DROP CRITERIA

ITEM	PACKAGED/UNPACKAGED	DROP DISTANCE	REQUIRED ACTION

Drop criteria must be specified for major components of the missile system.

3.6.3.1.2 Inspection or damage criteria. Inspection or damage criteria shall be preceded by a glossary of damage/inspection terms in the following format:

TERM	DEFINITION

Source: https://assist.dla.mil - Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

Inspection or damage requirements shall be in the following format:

ITEM	INSPECTION	DISPOSITION
	(Inspect for following)	
	(DEFECTS)	

3.6.3.2 Checkout. Checkout or functional checks shall provide the procedures for determining the operational suitability of the missile, its components and systems.

3.6.4 Ground handling and servicing. Ground handling and servicing contains instructions and illustrations, in the most logical order that are essential for:

a. Hoisting and moving procedures, giving points on the missile that may be used for ground handling, push points, steps, grips, etc, and the attachment and method of handling the complete missile by means of the hoisting sling. All hoisting slings, jacks, maintenance stands and platforms, jigs, towing bars, and mooring equipment shown shall be confined to items that are available to the operating unit. Positioning, closed height, and lift height required for all jacks from fully compressed shock struts and flat tires to extended shock struts and 2-inch tire clearance shall be given. The load capacity of all handling equipment required shall be specified. Dimensions of mooring attachments and methods of use shall be shown.

b. Shoring or counterbalancing the missile to remove and install the motor/engine.

c. Placing the missile in position upon the fixed or mobile ground launcher, placing it in live storage, or in alert configuration.

d. Servicing the missile with fuel and oil, and providing information regarding capacities, pressures, types of servicing fluids, etc., as applicable. The specific location of each filler cap and each servicing point shall be given.

e. Lubricating requirements, consistent with MIL-STD-838, and diagrammed as necessary for complete understanding.

f. Attaching rocket motor, booster units, Jet Assisted Take Off (JATO) equipment, special wiring connections, etc.

g. Leveling the missile transversely, and longitudinally, if applicable, including specific points where the leveling instrument shall be placed during this operation.

h. Covering other procedures necessary to prepare the missile for operation.

i. Define storage buildings, explosive class, etc. (Ref AFR 127-100).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

3.7 Chapter 4, inspection requirements. This chapter shall contain procedures to ensure serviceability of components for missiles maintained in storage or issued to fill requirements of using organizations. Tables may be used to augment procedures including correctable defects, non-correctable defects, etc. Procedures shall consist of receiving inspection, periodic inspection, scheduled inspections, and special inspections.

3.7.1 Receiving inspection. Receiving inspection shall contain procedures for inspection of inner package, and identification of contents.

3.7.2 Periodic inspection. Periodic inspection shall contain procedures using sampling procedures; inspection will consist of usual inspection for defects, electrical test of rocket motors, etc. Defects shall be separated into critical defects, major defects and minor defects. Periodic inspection shall also contain number of allowable defects, how to handle and dispose of rejected items and sampling charts.

3.7.3 Scheduled and special inspections. This is a complete inspection and operational test of missile and shall be scheduled at periodic intervals. Included shall be wing rolleron operational checks, component inspections, guidance and control section inspections, canard storage, wing storage, training missile, Nondestructive (NDI) inspection of training components and any other pertinent data required. Include missile inspection criteria and intervals if a missile is stored in an all up configuration.

3.8 Chapter 5, maintenance. This chapter shall contain essential instructions and illustrations for maintenance by the organizations which operate the missile. Maintenance shall be limited to repair, testing, replacement, and adjustments which can be performed at the user level. Repairs shall be confined to that work which can be performed without disassembly, or with partial disassembly, of the major components and systems which do not require the use of specialized shop test and calibration equipment. Removal of the components shall be included. Instructions shall cover each system of the missile, major components, assembly, or unit thereof. Instructions shall fully cover all applicable phases before proceeding to the maintenance of another system, component, assembly, or unit. Interdependence or other interrelationships shall, however, be acknowledged and explained. Instructions shall be concise and complete as to all essential information.

3.8.1 Test data. Instructions shall provide procedures for determining operational suitability of the missile, its components and systems, and the extent of maintenance required. A chart providing adjustment procedures and minimum performance standards shall be included. This shall outline indications, readings, and tests which will indicate whether a component, equipment, or system will provide acceptable results. This chart shall provide the means of establishing beyond reasonable doubt that the equipment is functioning properly, or in the case of an improper indication, it shall localize the difficulty to a component, assembly, or subassembly, designed for easy replacement. Illustrations shall be furnished which will enable clear visual identification of the means of connection between the test equipment and the equipment being tested.

Source: https://assist.dla.mil -- Downloaded: 202610-17T04:23Z
Check the source to verify that this is the current version before use.

3.8.2 Troubleshooting. Troubleshooting data shall be to a depth to support the maintenance concept (fault isolation to field level replaceable component). A list of the common failures that are likely to occur during checkout of the missile, a statement of probably causes of failures, and a satisfactory remedy for each failure shall be included. The list shall be divided into three columns.

Fault	Probable Cause	Remedy

3.8.3 Component removal. All steps required to detach and remove components, or systems, together with precautions to be observed, shall be included.

3.8.4 Disassembly. All steps required to dismantle, or take apart, the units shall be given.

3.8.5 Cleaning. Instructions shall be included for any cleaning methods required, noting the precautions to be observed. The Government specification numbers shall be given for any cleaning agents required.

3.8.6 Repairs and parts replacement. Procedures for minor repairs and parts replacement shall be given.

3.8.7 Reassembly. All steps for reassembly, mounting, or installing components of the systems shall be given, noting the precautions to be observed.

3.8.8 Checkout. Checkout data shall be given for the following functions in a step-by-step procedure.

a. Providing power to the system. Complete instructions shall be given for starting the generating system and connecting the power supply to the control panels, missile, and launching device. Proper means of establishing and maintaining correct power shall be explained. Instructions shall be provided for checking all circuits for proper connection and functioning, including warnings and cautions as required.

b. Starting missile guidance system. Instructions shall be provided for checking all related circuits for proper connection and function. Procedures for starting all gyros, radio, radar, electrical equipment, etc., employed in the operation of the guidance and tracking system, shall be given. Special emphasis shall be given to components which have specified warm-up period before they become effective. This warm-up period shall be stated in minutes.

c. Positioning of launching device. Complete instructions shall be given on procedures to be followed in setting azimuth and elevation angles. If the launcher must be "aimed", instructions for aiming shall be included.

d. Starting propulsion system. For those missiles with propulsion systems that must be started before launching, instructions for starting shall be provided.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

e. Adjustments. Instructions for regulating, rigging, or otherwise bringing to a correct relative position, including applicable tolerances and rigging check dimensions, shall be given.

3.8.9 Extreme weather maintenance. Instructions for maintenance required under conditions of extreme temperature range, dust, or humidity, shall be included.

3.8.10 Arming instructions. Rocket motor arming instructions, if applicable, shall be included.

3.8.11 Additional requirements. The following requirements apply to electrical and electronic equipment, and are in addition to (not in lieu of), the preceding items:

a. Electrical system. The general electrical characteristics, such as voltage rating, etc., shall be given. There shall also be a detailed ground operational check procedure for the electrical power supply system.

b. Electronic system. All of the equipment items making up the complete electronic installation shall be listed, giving the function of each item and the model or type designation. There shall also be a detailed ground operational check procedure for the electronic system.

3.8.12 Illustrations. The following are desirable illustrations that should be included in Chapter 5 with additions and omissions permitted as applicable to a particular missile.

a. Access and inspection openings diagrams, indicated by index number and tabulation of the equipment or parts for which access is provided.

b. Lubrication charts or diagrams.

c. Surface control system diagrams, (separate diagrams may be furnished for surface control locks.)

d. Wing flap control system diagram.

e. Speed dive brake system diagram.

f. Engine mount. (An enlarged view of provisions for vibration absorption, if used.)

g. Assisted takeoff installation.

h. Main landing gear installation, wheel, ski, or float types (including water rudder and control system).

i. Recovery systems (chutes, shock absorbing, flotation devices, etc).

j. Hydraulic systems diagrams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

k. Heating, ventilating, and refrigeration systems diagrams.

l. Pressurization system diagram.

m. De-icing system diagram.

n. Engines and engine accessories installation.

o. Engine adapting parts diagram. (This illustration shall show the complete engine ready for installation with all special adapting parts. These parts shall be shaded, or patterned, to identify them as parts that are not received with a replacement engine and must be removed from the old engine for installation on the new engine.)

p. Fuel system diagram.

q. Purging or fuel vapor dilution systems diagram (schematic).

r. Oil system diagram (schematic), including oil dilution system.

s. Bomb racks, internal and external, including sway bracing.

t. Chemical tank installation.

u. Photographic system installation.

v. Main landing gear actuating system diagram.

w. Engine water injection system (schematic).

3.8.13 Wiring diagrams. Include wiring diagrams and associated data necessary for electrical and electronic maintenance technicians to: understand the makeup and function of each power, control, and signal circuit; trace each circuit; make continuity checks and accomplish general and specific troubleshooting. It shall correlate its working relationship with material in other sections pertaining to electrical or electronic items, how to locate a particular diagram, an explanation of the wiring identification (numbering) system used on the diagrams, and any other information considered essential for proper understanding and use. This shall consist of a list of the titles of all wiring diagrams arranged in alphabetical order. Circuit diagrams shall be provided for each electrical equipment item or system and for each electronic item or system. The diagrams shall show all cables and related accessories used for interconnection of equipment. Location of equipment and accessories, routing of cables, and connection of cables shall be shown to the extent necessary for an understanding of the circuit. Complete internal wiring of equipment shall not be shown. Each diagram shall be assigned a figure number and title. Diagrams and the graphic symbols used shall conform to the standards cited in MIL-M-38784. A "Table of Critical Wire Length" shall be included on a diagram, when necessary, to indicate the maximum allowable length for a given size of wire in circuits handling heavy

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

current loadings, balanced circuits, etc. Wires shall be identified by numbers exactly as they are in the missile. The equipment index number shall be shown immediately adjacent to the related equipment item. Any circuits left in "mid-air" due to being picked up on other diagrams shall be labeled as a part of the circuit to which they are connected. Within existing space limitations, the diagrams shall show items in their true positions relative to one another. This is particularly important in the case of terminal arrangement.

3.9 Chapter 6, packing, shipping and storage. The chapter shall provide procedures for packing the missile components in the appropriate containers, including shipping and storage instructions.

3.9.1 Packing. Packing will be accomplished per special packaging instructions (SPI) and reference shall be made to the applicable SPI. Components shall be packed in original containers, when possible, utilizing shipping plugs/caps as required.

3.9.2 Shipping and storage. Shipping and storage shall be per existing regulations. Empty shipping containers and dunnage should be retained for future use.

3.9.3 Shipping and storage containers. Explosive components of the missile are packed per TO 00-85 series and shall be referenced. Show all shipping and storage containers in tabular format. Receiving inspections will be performed per existing directives, upon receipt of missile components from manufacturer, depot or using organization. Show table listing all applicable inspection requirements. Containers will be marked per existing directives.

3.10 Chapter 7, contingency procedures. This chapter may contain instructions pertaining to special flight evaluation configurations, such as installation of a telemetering unit, and other procedures may be required. This chapter will also contain step-by-step procedures for each emergency condition that may occur during assembly, handling, servicing, or maintenance, and instructions for assembly and inspection at a forward base under heavy engagement.

3.11 Chapter 8, training missiles. This chapter shall contain instructions and procedures to ensure serviceability of training missiles. Instructions and procedures will normally include the same information as required by 3.3.1.1, 3.6, 3.7, 3.8 and 3.9. Training missiles should be designated per MIL-STD-875.

3.12 Chapter 9, illustrated parts breakdown. This chapter shall consist of an illustrated parts breakdown in accordance with MIL-M-38807. Manuals containing an illustrated parts breakdown shall not exceed 800 pages. A separate illustrated parts breakdown shall be acquired for manuals exceeding 800 pages.

3.13 Chapter 10, difference data. This chapter will contain all difference data on a particular model or version of missile. This chapter will be arranged like a complete mini-manual, including all pertinent sections, including Illustrated Parts Breakdown. NOTE: A numerical index shall be included if the Maintenance Parts List contains more than 25 items.

4. QUALITY ASSURANCE PROVISIONS

Source: https://assist.dla.mil -- Downloaded: 202610-17T04:23Z
Check the source to verify that this is the current version before use.

4.1 The quality assurance provisions shall be in accordance with MIL-M-38784 and MIL-P-38790.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging, packing, and marking for shipment shall be in accordance with MIL-M-38784 and MIL-P-38790.

6. NOTES

6.1 Intended use. Manuals prepared per the requirements of this specification are intended for field level maintenance of guided bombs, guided rockets, guided aircraft rocket interceptors, air-to-air missiles, air-to-surface missiles, and pilotless spacecraft.

6.2 Data requirements. When this specification is used in an acquisition which incorporates DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements.

Paragraph No.	Data Requirement Title	Applicable DID No.	Option
3.1	Technical Orders	DI-M-3407	Manual Technical: Assembly, servicing and field maintenance instructions, with illustrated parts breakdown.

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 16

Preparing Activity
Air Force - 16

(Project TMSS - F476)

Review Activities:
Air Force - 01, 10, 99

User Activities:
Air Force - 11, 18, 70, 84, 71, 82

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

NSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in ubmitting comments and suggestions for improvements. All users of military standardization documents are invited to provide uggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was oo rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wordin; changes which would alleviate the roblems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an cknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being onsidered.

IOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of pecification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization o waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

RTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

FICIAL BUSINESS
NALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

HQ AFLC/LOLME
Wright-Patterson AFB OH 45433

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil – Downloaded: 2026-10-17T04:23Z
Check the source to verify that this is the current version before use.